

Lab 6 - Twirly bob

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Learning Outcomes

- Collaborate with other students in problem-solving and laboratory activities
- Demonstrate good scientific practices
- Analyze real-world experimental data
- Relate the results to the physical concepts discussed in the lecture
- Identify the forces acting on the masses in the system draw their FBD's
- Apply Newton's 1st, 2nd and 3rd law of motion to uniform circular motion
- Calculate the centripetal acceleration of an object in uniform circular motion
- Identify the centripetal force keeping an object in uniform circular motion

Safety awareness

This lab requires you to wear a hard hat and safety goggles.

Use common sense and be aware of your surroundings when working with the experimental apparatuses.

Make sure no one is in danger of getting struck by the "centripetal force apparatus" during the experiment. π

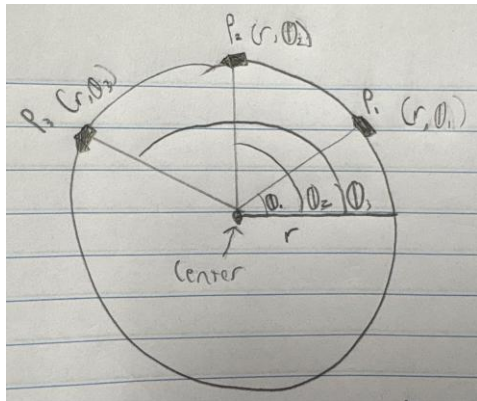
Material/Equipment

- ~~Tube~~
- ~~20g mass with a hole (twirling mass)~~
- ~~$\approx 1.5\text{m}$ of string~~
- ~~Slotted hook mass (hanging mass)~~
- ~~Tape~~
- ~~Hardhat~~
- ~~Safety goggles~~
- ~~Stopwatch~~
- ~~Meter stick~~

Prelab:

First, complete steps 1 and 2 individually in your notes. Then collaborate as a group to complete steps 1, 2 and 3 of the lab report.

1. Insert a sketch or drawing of a circle and the location of an object in circular motion. Label the center and radius of the circle and the position of the object at a 3 points in time using r and θ .



2. Write definitions, equations and units for the following uniform circular motion terms:

Uniform circular motion: Motion of an object moving at a constant speed while the direction is constantly changing

Radial position (r): $r = R$, distance from the center of the circle to the edge, units: meters

Angular position (ϕ): $= \frac{s}{r}$, the angle that shows the objects position on the circle relative to the starting point, units: radians

Arc length (**S**): = $r\phi$, distance traveled along the path,

units: meters

Angular velocity (**ω**): $\frac{\Delta\phi}{\Delta t} = \frac{v}{r}$, the rate of change for

angular position over time, units: radians/second

Circumference (**C**): $2\pi r$, the distance around a full

revolution, units: meters

Period (**P**): time required to complete a revolution. $\frac{2\pi}{\omega}$,

units: seconds

Tangential velocity (**v**): $v = r\omega$, linear speed of the

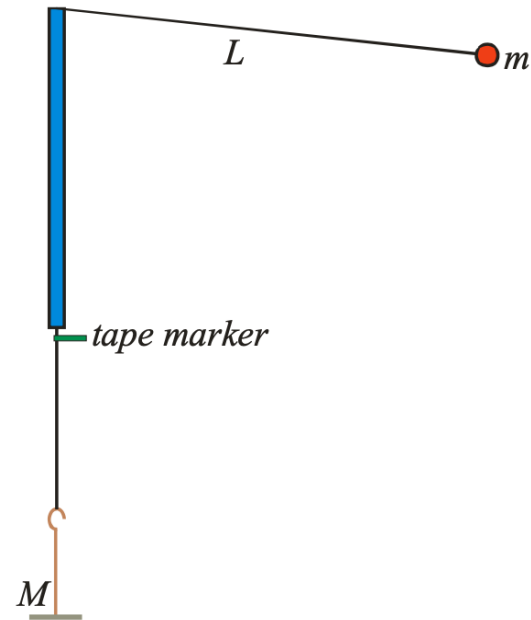
object, units: meters/second

Centripetal acceleration (**a_c**) = $\frac{v^2}{r} = r\omega^2$, acceleration

toward the center of the circle, meters/second²

Centripetal force (**F_c**) = $ma_c = \frac{mv^2}{r}$, the inward force keeping the object in a circular

motion, units: newtons



3. Check off all the materials/equipment required for the lab.

Setup

Identify the first 5 materials you checked off in the figure to the right.

1. Assemble the centripetal force apparatus as demonstrated by the instructor by performing the following steps:
 - a. Obtain a 1.5 m length of string.
 - b. Thread the string through the tube.
 - c. Attach a mass **m** to one end of the string. (e.g. **m** = 20 g to start). Knot and tape!
 - d. Attach a mass hanger to the opposite end of the string such that the mass hanger has a total mass of **M** and hangs straight down from the apparatus when the apparatus is vertical, as indicated in the diagram. (e.g. **M** = 50-70g to start or whatever mass makes for smooth, stable revolutions).

- e. Pull out $L = 0.50$ m of string on the m end of the apparatus such that m will be able to make up to a 0.50 m circular radius when whirled horizontally.
- f. Mark the part of the string about 1.0 cm immediately below the tube with a piece of tape. This will be a visual marker to keep the radius of the circular path constant.
 - 1.
 2. **Put on a hard hat and safety goggles. Then, one at a time, try** whirling m until you can keep the top of the tape marker about 1.0 cm from the bottom of the tube. When you whirl m , you will use this marker to determine the radius of the circle in which m traveled. If the marker rises or falls appreciably as m whirls, the radius of the circle is changing.

Hold the tube vertically and keep the circular path of m above your heads.

Practice! One of you may be better at this than the others. The tape should not travel up the tube or slide down from this position. If the marker touches the bottom of the tube, M no longer will be supplying the centripetal force!

You may hold the marker to get mass m in motion, but then you must let go

You may not hold the string during measurements!

Stop and Think

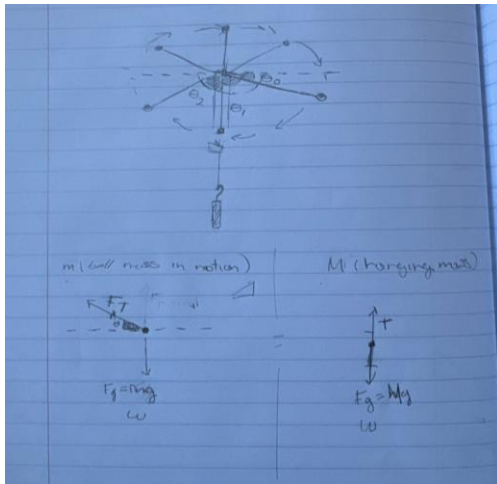
1. Does your group's stable rotation configuration match the figure above?
If so, label the angle θ from the horizontal in the figure.

The stable rotation does match the figure above.

2. **Why is it not possible to twirl the mass m perfectly horizontally (at an angle of $\theta = 0^\circ$)?** (Hint: questions 3-7 will help you to answer this question. Do them individually in your notes first, then collaborate before entering group answers).

The angle can't reach zero because then there would be no balancing tension force for the vertical axis

3. **Draw two free-body diagrams: one for system " m " and one for system " M " showing the forces acting on each. Neglect any friction opposing motion.**



4. Which force is independent of mass in both FBDs?

The tension is independent because adding mass to either object increases the tension of the whole system.

5. Which force is not independent of mass in the FBDs?

The gravity is not independent because adding mass to either object increases the tension of the whole system.

6. Write Newton's 2nd law equations for each mass separately. Remember, you will get one equation for each orthogonal direction.

For M (in y): $T - Mg = 0$

$$T = Mg$$

For m (in y): $T_y = T(\sin(\theta))$

$$T(\sin(\theta)) - mg = 0$$

$$T(\sin(\theta)) = mg$$

For m (in x): $T_x = T(\cos(\theta))$

$$T(\cos(\theta)) - ma_c = 0$$

$$T(\cos(\theta)) = ma_c$$

In which equation is F_{net} not equal to zero? $T(\cos(\theta)) = ma_c$

According to your FBD, is F_{net} a center-seeking force?

It is not a center-seeking force.

If the only force acting in the y-direction on m were $F_g = mg$, could you twirl the mass horizontally, so that the radius of the circular motion R is equal to L ?

No, because the only force would be gravity which points down and the mass would fall down

After discussing this in your group, go back and explain your answer to question 2 above.

OK

7. From the 2 y-equations in 6, derive an expression for the angle θ in terms of the masses m and M only.

$$\theta = \sin^{-1}\left(\frac{m}{M}\right)$$

8. Write the radius of circular motion, R , in terms of L and the angle θ .

$$R = L \sin(\theta)$$

Do not enter any values into the equations in 8 and 9. Leave the equations in terms of the variables (algebraic form) until you collect and analyse your data.

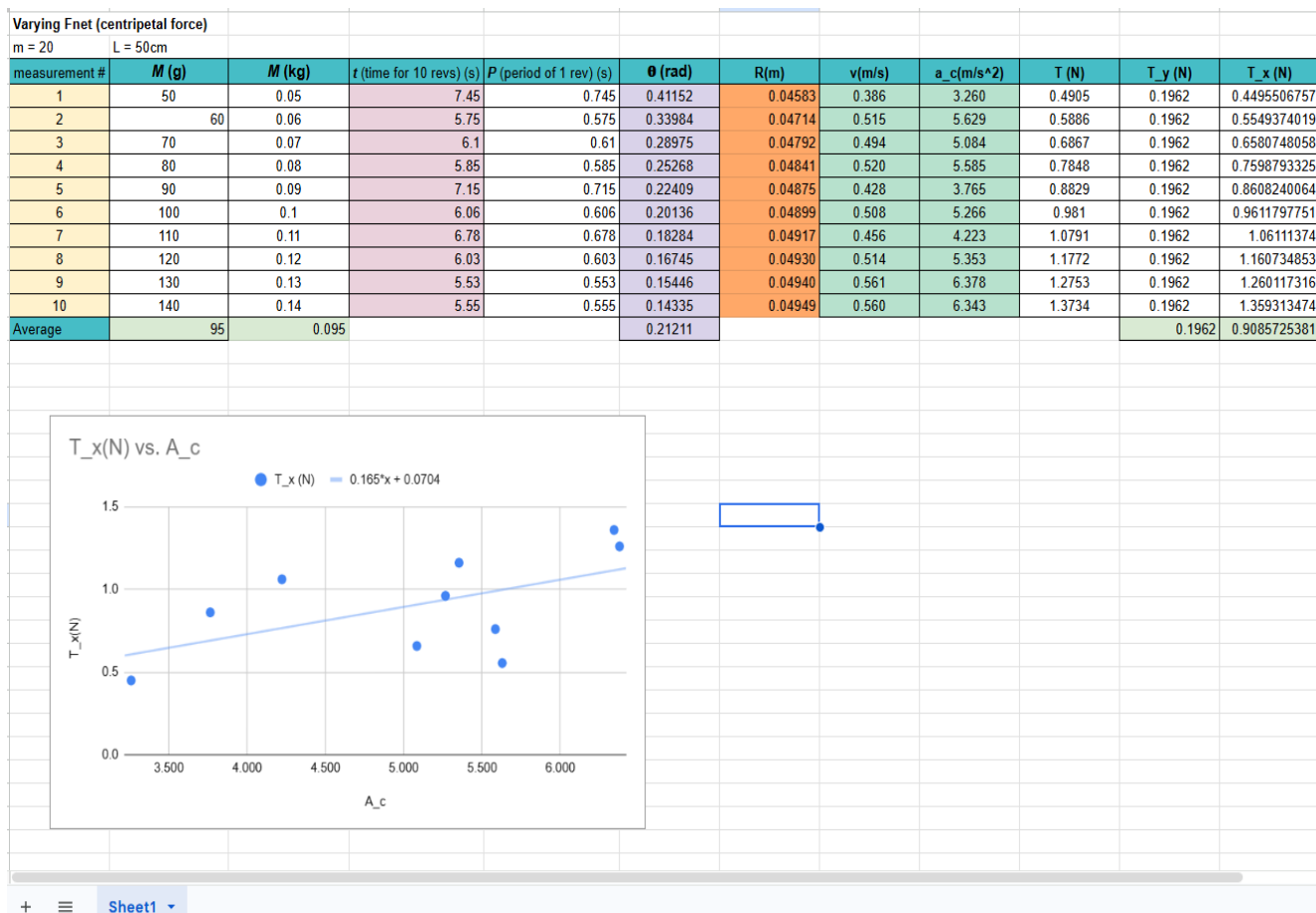
Take measurements (collect data)

You will need 3-4 people at a time to take and record data. One to keep mass m in uniform circular motion, one to operate the stopwatch, one to count 10 full revolutions, and one to enter measurements into the [Lab 6 spreadsheet](#) (copy template).

Varying F_{net}

1. Use a stopwatch to **measure the time it takes for 10 revolutions**. Record the time t and the value of M_o in the spreadsheet.
2. Use a cell formula to **calculate the period P for 1 revolution** as $t/10$. Copy and paste this formula for the entire period column.
3. Increasing the mass of the hanger in 10 g increments up to $M_o + 100$ g (or increments of 5g, or 15g, as long as you have 10 data points). **For each mass M , measure the time t for 10 revolutions**. Record t , M and P in the spreadsheet (P should populate automatically according to the cell formula).
4. Use a cell formula and equation 8 in Stop and Think to **calculate θ** . Copy and paste this formula for the theta column.
5. Use a cell formula and equation 9 in Stop and Think to **calculate R** . Copy and paste this formula for the radius column.
6. Use a cell formula and the prelab equation for tangential velocity to **calculate v** . Copy and paste this formula for the velocity column.
7. Use a cell formula and the prelab equation for centripetal acceleration to **calculate a_c** . Copy and paste this formula for the acceleration column.
8. Use a cell formula and the equation “For M (in y)” to **calculate the tension in the string**. Copy and paste this formula for the tension column.
9. Use a cell formula and “For m (in y)” to **calculate the y-component of tension in the string**. Copy and paste this formula for the tension column.
10. Use a cell formula and “For m (in x)” to **calculate the x-component of tension in the string**. Copy and paste this formula for the tension column.

Insert a screenshot of your data table and a link to the googlesheet.



Data Analysis

Based on Newton's 2nd Law equations in Stop and Think...

1. Which column in the spreadsheet should be equal to the weight of the hanging mass? Calculate the %-difference and explain why it is not 0.

The weight of M should equal mg, because $T_y = mg$

Since the string is not perfectly vertical, the Force_T > Mg

$$\% \text{ difference} = |T_y - Mg| / Mg \times 100\%$$

$$= |0.1962 - 0.196| / 0.196 \times 100\%$$

$$= 0.0001\% = 0\%$$

$$T_y = MG$$

2. Which column in the spreadsheet should be equal to the weight of the twirling mass? Calculate the %-difference and explain why it is not 0.

The column that should equal the weight is $T_y(N)$

The percent difference is not exactly 0 because the radius changes slightly due to uncertainty and human error.

$$\% \text{ difference} = |T_y - mg| / mg \times 100\%$$

$$= |0.1962 - 0.196| / 0.196 \times 100\% = 0.0001\% = 0\%$$

3. Which column in the spreadsheet should be equal to the centripetal force F ?

The T_x column would represent the centripetal force F .

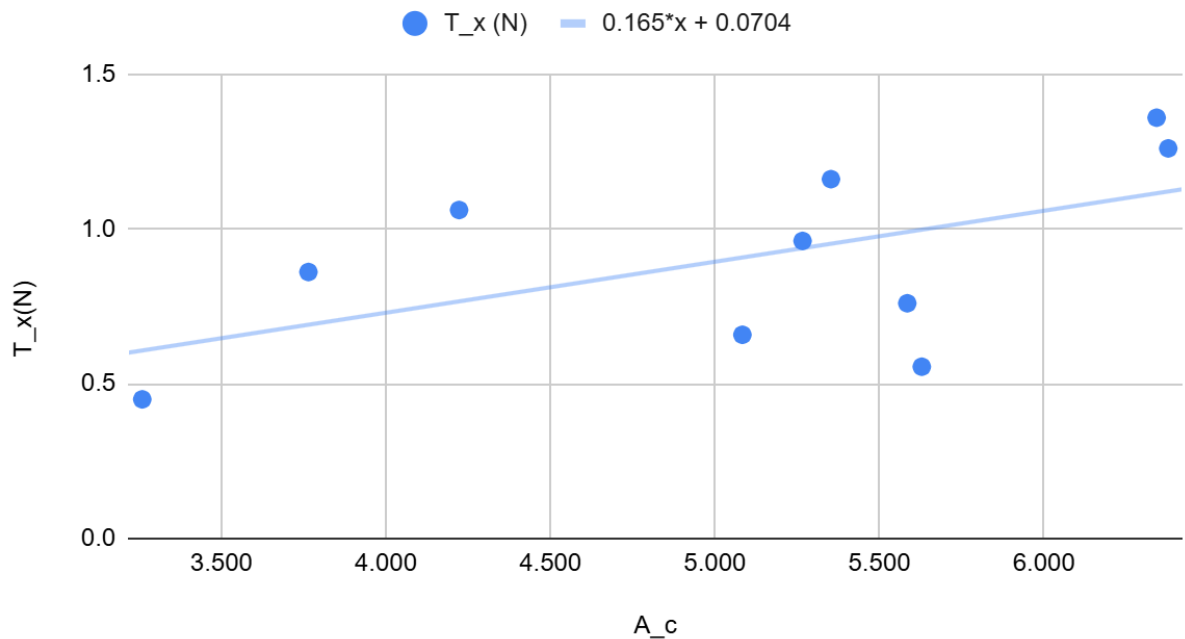
According to Newton's 2nd Law, a net (unbalanced) force is required to change an object's velocity and keep it accelerating. For an object moving in uniform circular motion (at a constant speed), the centripetal force should be directly proportional to the centripetal acceleration. You will now do some graphing analysis to check this.

4. Write $F_c = ma_c$ and the equation "For m (in x)" from Stop and Think.

$$F_c = T(\cos(\theta)) = ma_c$$

5. Plot T_x versus a_c and find the best fit straight line.
- Select the columns for centripetal acceleration and the x-component of tension in the spreadsheet.
 - Insert a chart that is a scatter plot.
 - Do a best linear fit and display the trendline equation on the chart.

$T_x(N)$ vs. A_c



6. Identify the slope of this line. What physical quantity does it represent (quantity and units).

$M = 0.165$ This represents the mass (with uncertainty)

7. How does your slope compare to your expectation? Calculate a %-difference and explain why it is not 0.

The slope meets expectations when you factor out the uncertainty.

Expectations: $mg = 0.02\text{kg}(9.8\text{m/s}^2) = 0.1962\text{N}$

% difference = $|0.1962 - 0.1960| / 0.1960 \times 100\% = 0.001\%$

It may not be 0% due to experimental error but the % difference is actually 0 % with significant figures.

According to Newton's 3rd Law, for a pair of objects interacting, the force object A exerts on B is equal and opposite to the force object B exerts on A. Identify the force pair and type of force responsible for uniform circular motion.

The force pair is the force of the twirling mass m and the force of the mass M . These forces together lead to the force of Tension. This causes circular motion according to Newton's 3rd Law because the forces are equal in magnitude and opposite in direction.

The force from the twirling mass is a centripetal force that provides constant acceleration towards the center of the system. This tension force equals the weight or force of gravity of the mass M .

Take aways

How did your group meet the Learning Objectives of this lab?

We collaborated with other students to plan out how we would do our trials, we asked other groups for their take on the trails and how they calculated their answers. We practiced good scientific practices by ensuring we limited sources of error as much as we could. We collected data we got from the experiment and recorded it. We also used Newton's laws to calculate the numbers using the collected data.

Which Learning Objective did you find the most impactful and why? Each group member should answer for themselves.

Justas Radavicius: The most impactful learning objective was collaborating with other students because it shows that science is all about working together.

Jared Lomeli: "Identify the centripetal force keeping an object in uniform circular motion" because this was our first lab observing centripetal force and turning it into usable data.

Juanita Taborda Arias: Use the second Newton Laws for each mass separately, this give me a better understanding

Ethan Dickson-Child: I thought the objective of analysing real world experimental data was the most impactful. This is because seeing odd changes in recorded data forced me to think about what exactly is happening in the experiment in order to find the cause of the outliers.

How could your group most improve this lab? This may be a collaborative answer.

This lab could be improved by using something related with capstone and possibly a sensor so that we could record Newton's and other important data. To remove the errors in the spinning weight from going down, we could utilize a board that keeps the mass horizontal; however this would introduce friction to the spinning weight.

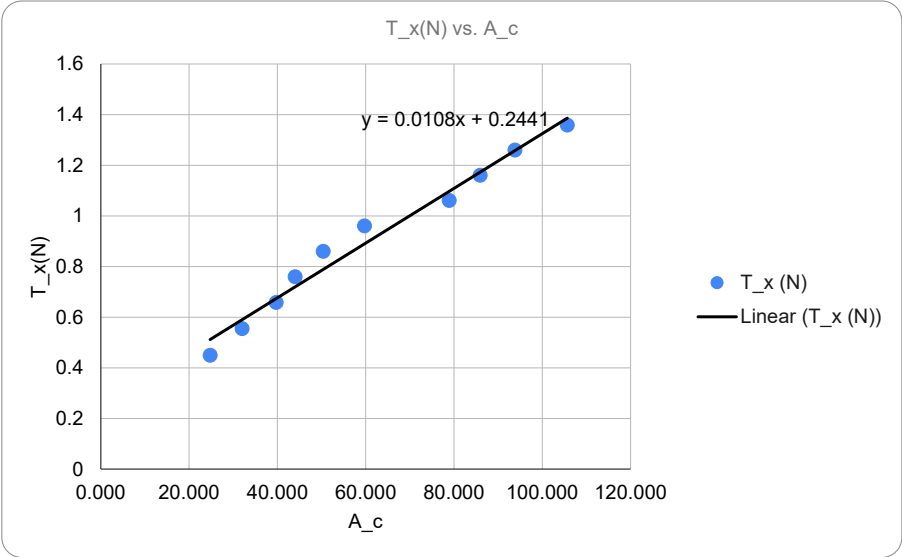
Varying Fnet (centripetal force)

m = 20 L = 50cm

corrected corrected

when L = 50 corrected corrected

measurement #	M (g)	M (kg)	t (time for 10 revs) (s)	P (period of 1 rev) (s)	θ (rad)	R(m)	v(m/s)	a_c(m/s^2)	T (N)	T_y (N)	T_x (N)
1	50	0.05	8.55	0.855	0.41152	0.45826	3.368	24.748	0.4905	0.1962	0.449550676
2	60	0.06	7.63	0.763	0.33984	0.47140	3.882	31.967	0.5886	0.1962	0.554937402
3	70	0.07	6.9	0.69	0.28975	0.47916	4.363	39.732	0.6867	0.1962	0.658074806
4	80	0.08	6.59	0.659	0.25268	0.48412	4.616	44.009	0.7848	0.1962	0.759879333
5	90	0.09	6.18	0.618	0.22409	0.48750	4.956	50.391	0.8829	0.1962	0.860824006
6	100	0.1	5.69	0.569	0.20136	0.48990	5.410	59.737	0.981	0.1962	0.961179775
7	110	0.11	4.96	0.496	0.18284	0.49167	6.228	78.898	1.0791	0.1962	1.06111374
8	120	0.12	4.76	0.476	0.16745	0.49301	6.508	85.901	1.1772	0.1962	1.160734853
9	130	0.13	4.56	0.456	0.15446	0.49405	6.807	93.799	1.2753	0.1962	1.260117316
10	140	0.14	4.3	0.43	0.14335	0.49487	7.231	105.661	1.3734	0.1962	1.359313474
Average	95	0.095			0.21211					0.1962	0.908572538



original data for time column

7.45
5.75
6.1
5.85
7.15
6.06
6.78
6.03
5.53
5.55